

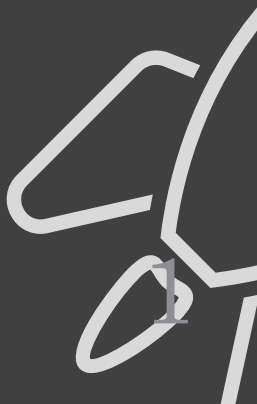


IBM Data Science Capstone Project

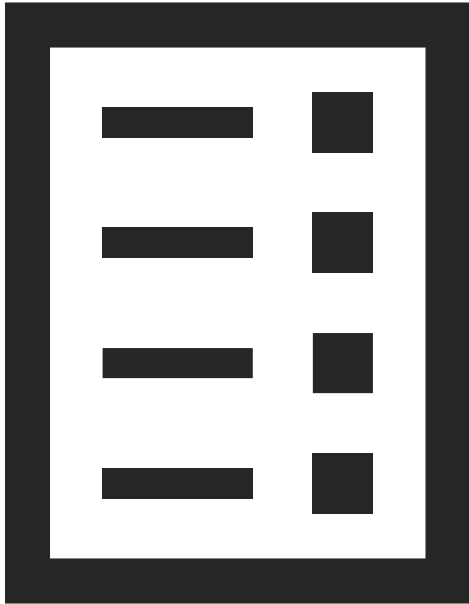
Md Muktadirul Islam Jafir

<https://github.com/mmi-jafir/IBM-Data-Science-Capstone-Project>

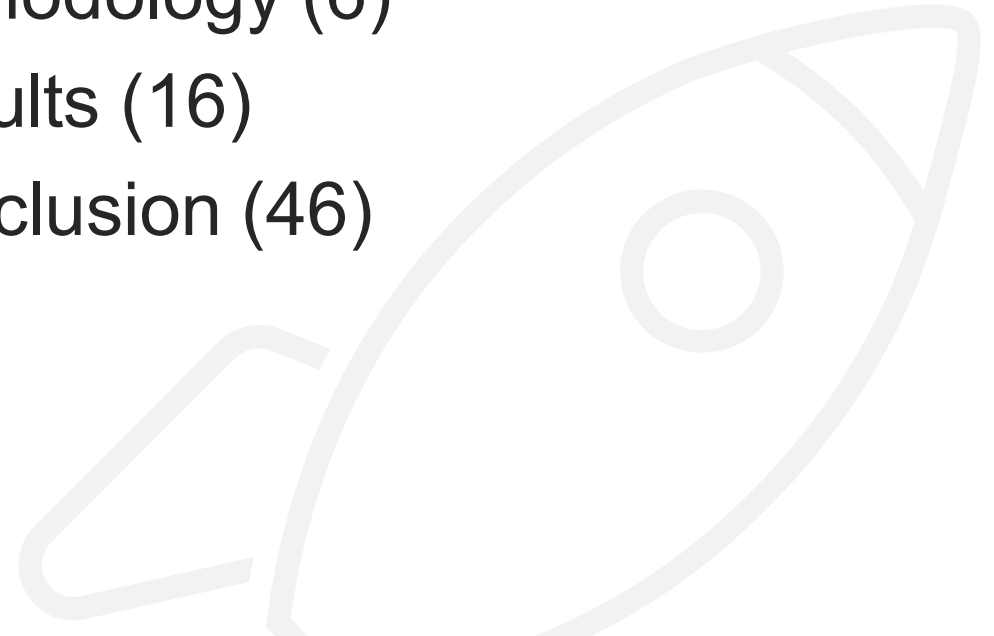
15/05/2026



Outline



- Executive Summary (3)
- Introduction (4)
- Methodology (6)
- Results (16)
- Conclusion (46)



Executive Summary

- Data was collected from the public **SpaceX API** and **Wikipedia launch records** to analyze Falcon 9 mission outcomes. A target variable (class) was created to classify successful first-stage landings. The dataset was explored using **SQL analysis, visualizations, interactive maps, and dashboards** to identify trends and relationships.
- Relevant features were selected for predictive modeling, with categorical variables transformed using **one-hot encoding** and numerical features standardized. **GridSearchCV** was applied to optimize model parameters.
- Four machine learning models were developed: **Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors**. All models achieved similar performance, with an accuracy of approximately **83.3%**. While the models showed strong predictive capability, they tended to overestimate successful landings, indicating that additional data could further improve model accuracy and reliability.

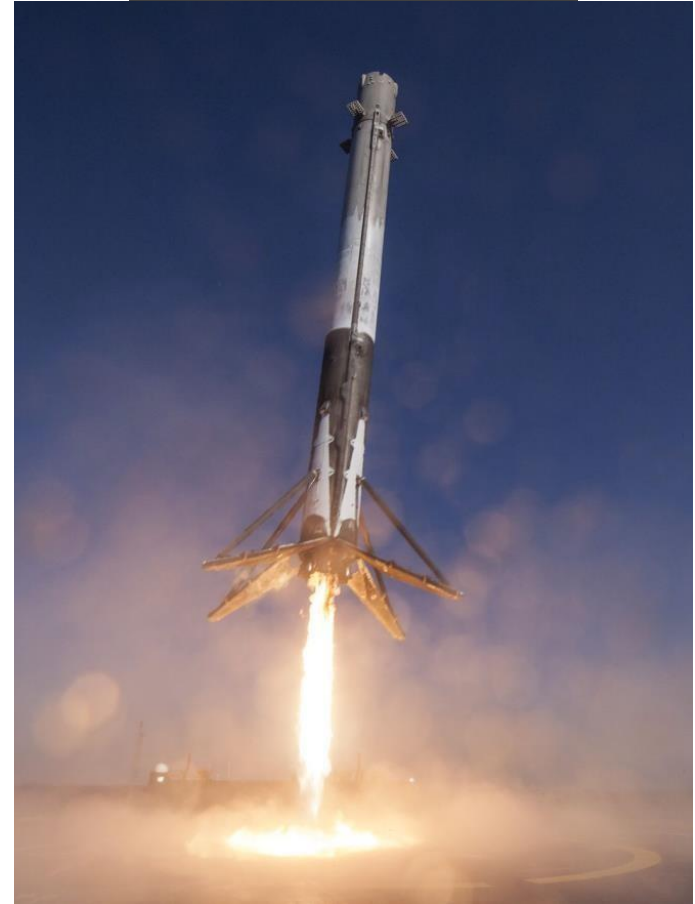
Introduction

Background

- The commercial space industry has entered a new era of rapid growth and competition.
- **SpaceX has significantly reduced launch costs**, with Falcon 9 launches costing approximately **\$62 million compared to around \$165 million for competitors**.
- A major factor behind this cost advantage is the successful **recovery and reuse of the first-stage booster**.
- A competing company, **Space Y**, aims to challenge SpaceX in this market.

Problem Statement

Space Y requires a **machine learning model to predict the likelihood of successful first-stage booster recovery**, enabling better cost estimation and strategic decision-making for future launches.

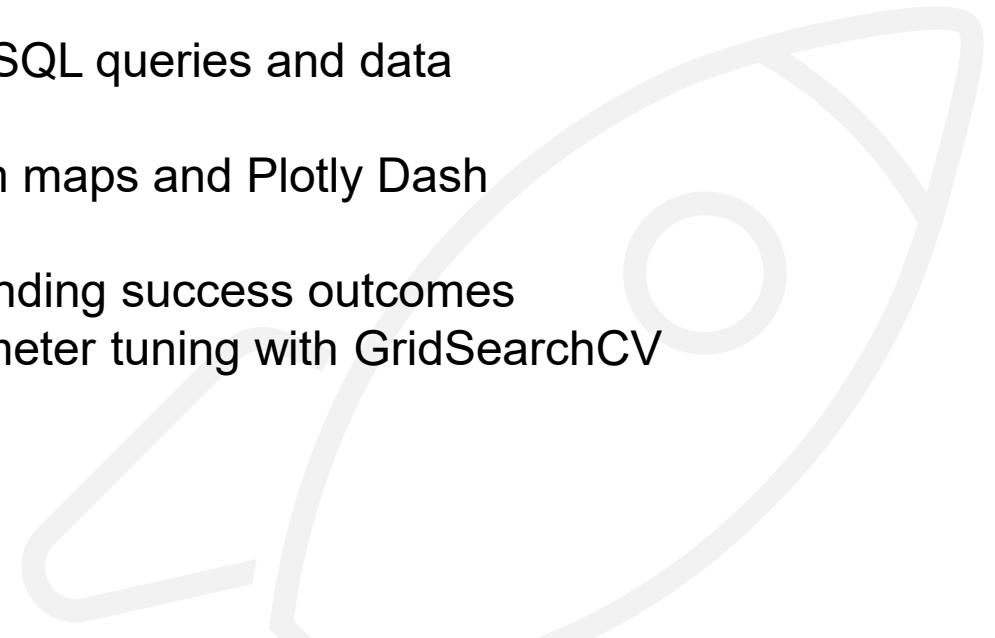


SpaceX Falcon 9 Rocket – The Verge

Methodology

Data Collection & Analysis Methodology

- Collected launch data from the SpaceX public API and SpaceX Wikipedia launch records
- Cleaned and prepared the dataset through data wrangling and preprocessing
- Created a target variable by classifying successful first-stage landings as successful and all others as unsuccessful
- Conducted exploratory data analysis (EDA) using SQL queries and data visualizations to identify trends and relationships
- Developed interactive visual analytics using Folium maps and Plotly Dash dashboards
- Built predictive classification models to estimate landing success outcomes
- Optimized model performance through hyperparameter tuning with GridSearchCV



Methodology

Overview of Data Collection, Wrangling, Visualization, Dashboard,
and Model Methods

Data Collection Overview

The data collection process combined **API-based data extraction** from the public SpaceX API with **web scraping of launch records** from the SpaceX Wikipedia page.

The following slides present the workflow for each data source:

Next slide: Data collection process using the SpaceX API

Following slide: Data collection process using web scraping from Wikipedia

SpaceX API data fields included:

Flight Number, Date, Booster Version, Payload Mass, Orbit, Launch Site, Outcome, Flights, Grid Fins, Reused, Legs, Landing Pad, Block, Reused Count, Serial, Longitude, and Latitude.

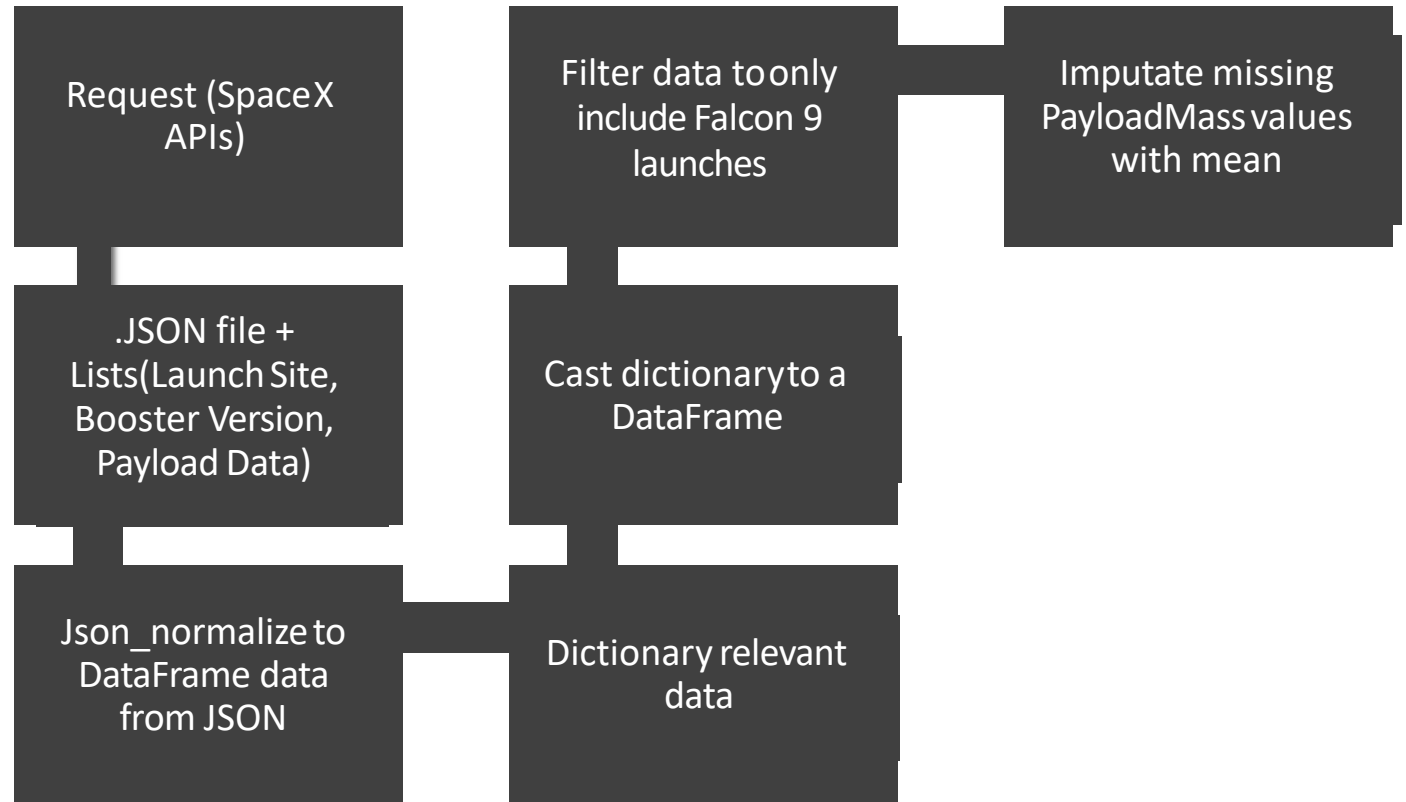
Wikipedia web-scraped fields included:

Flight Number, Launch Site, Payload, Payload Mass, Orbit, Customer, Launch Outcome, Booster Version, Booster Landing, Date, and Time.

Data Collection – SpaceX API

GitHub url:

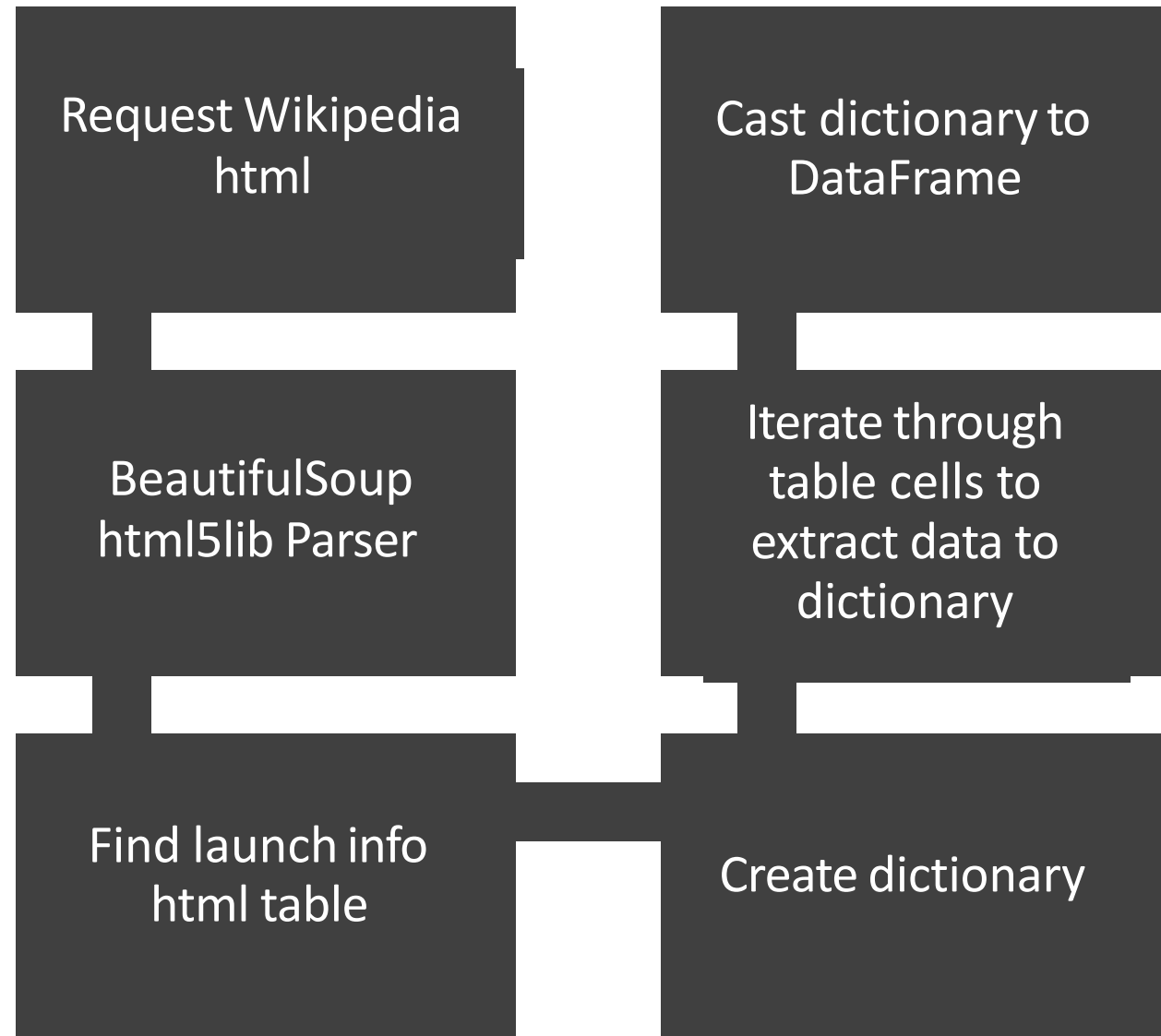
<https://github.com/mmi-jafir/IBM-Data-Science-Capstone-Project>



Data Collection – Web Scrapping

GitHub url:

<https://github.com/mmi-jafir/IBM-Data-Science-Capstone-Project>



Data Wrangling

Target Variable Creation

A training label was created to classify first-stage landing outcomes, where **successful landings were assigned a value of 1** and **failed landings a value of 0**.

The original **Outcome** field contained two components:

- **Mission Outcome**
- **Landing Location**

A new target variable, **class**, was generated based on mission success:

Mapped as Successful (1):

True ASDS

True RTLS

True Ocean

Mapped as Unsuccessful (0):

None None

False ASDS

None ASDS

False Ocean

False RTLS

EDA with Data Visualization

Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted on key variables including **Flight Number**, **Payload Mass**, **Launch Site**, **Orbit**, **Landing Outcome (class)**, and **Year**.

Visualizations included:

- Flight Number vs. Payload Mass
- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Orbit vs. Success Rate
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Success Trend

Scatter plots, line charts, and bar charts were used to examine patterns and relationships between variables, helping identify meaningful features for machine learning model training.

EDA with SQL

SQL Analysis

The dataset was loaded into an **IBM DB2 database** and queried using **SQL through Python integration**.

SQL analysis was performed to gain deeper insights into the dataset, including information on:

- Launch site names
- Mission outcomes
- Payload sizes across different customers
- Booster versions
- Landing outcomes

These queries helped improve understanding of launch performance patterns and operational characteristics.

Build an interactive map with Folium

Geospatial Analysis

Folium maps were used to visualize launch sites, successful and unsuccessful landing outcomes, and proximity to key geographic features such as **railways, highways, coastlines, and nearby cities**.

This geospatial analysis helps explain the strategic placement of launch sites and provides insight into how location may influence landing success.

Build a Dashboard with Plotly Dash

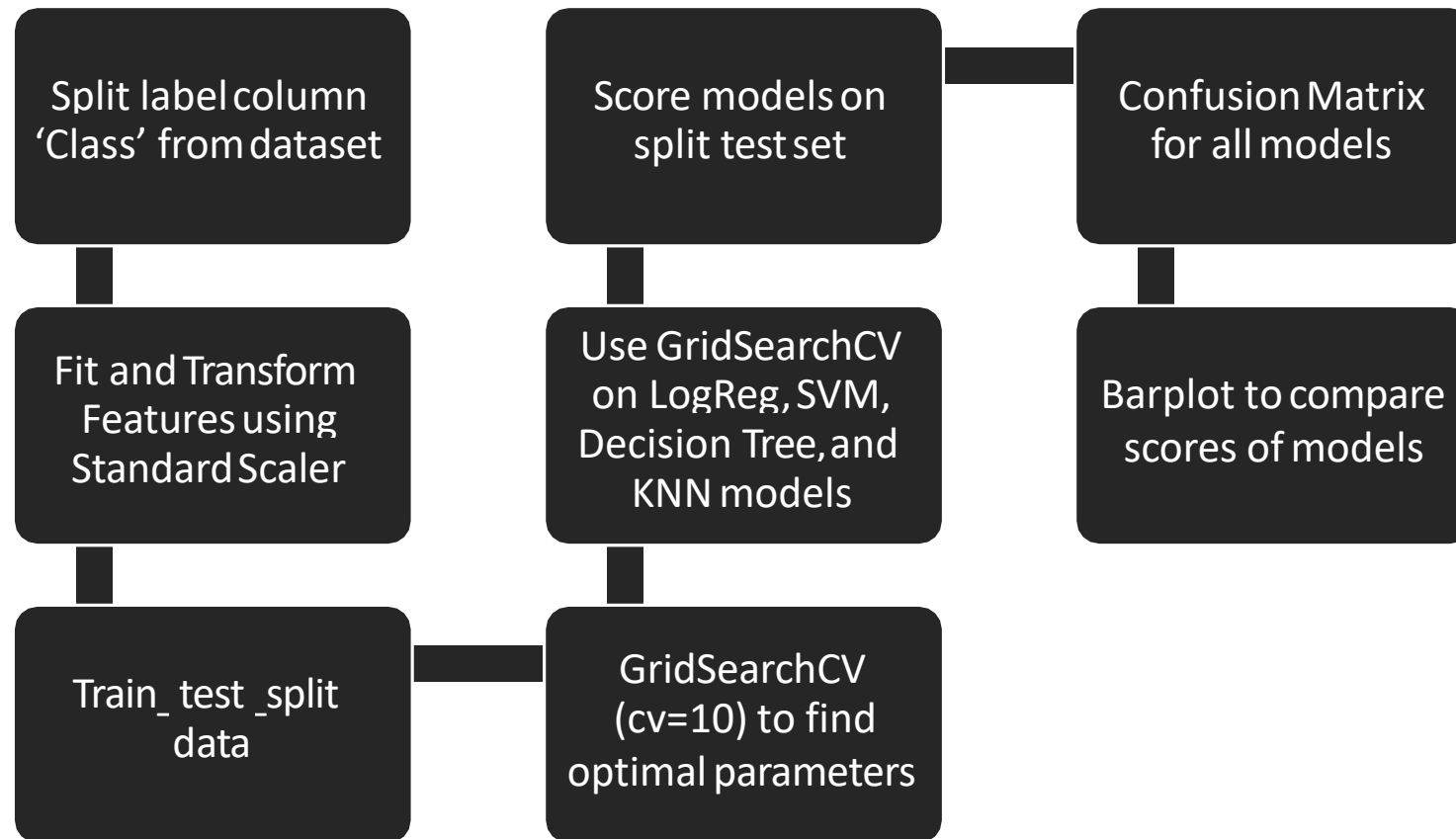
Interactive Dashboard

The dashboard includes an interactive pie chart and scatter plot for launch performance analysis.

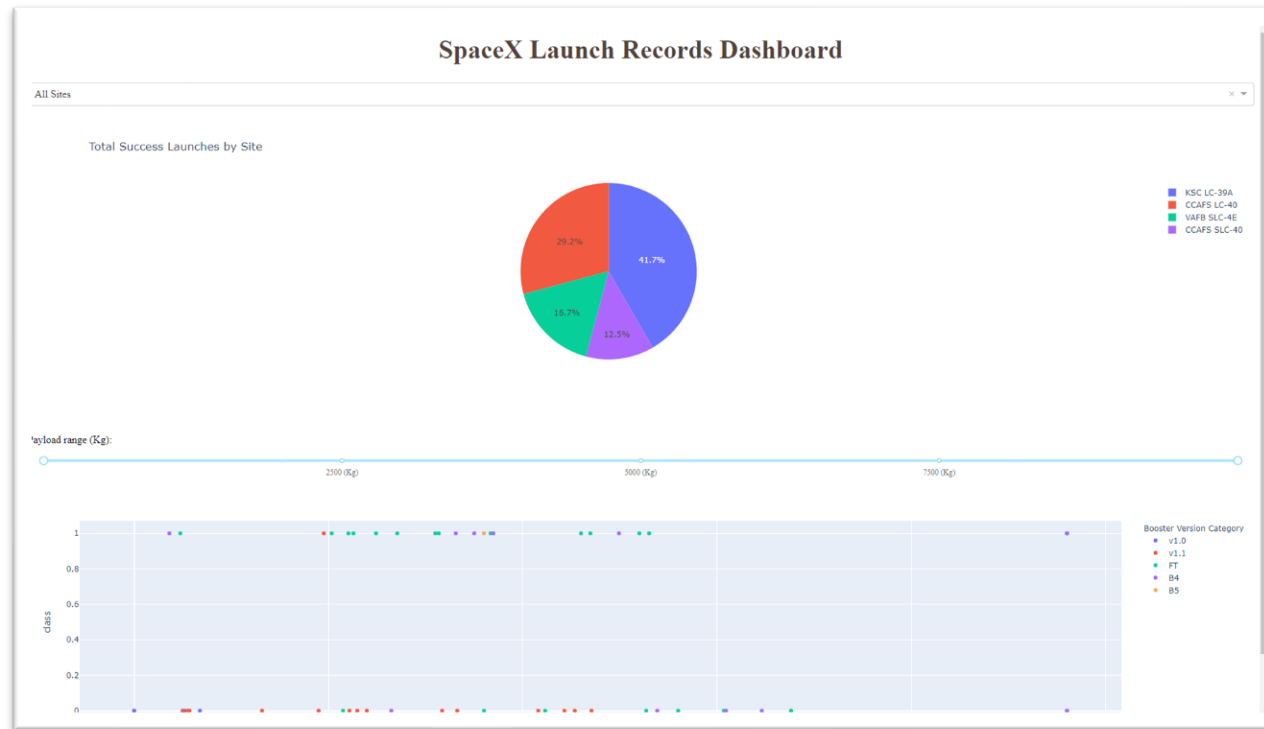
- The pie chart displays the distribution of successful landings across all launch sites or the success rate for a selected individual launch site.
- The scatter plot allows filtering by all launch sites or a specific site, with a payload mass slider ranging from 0 to 10,000 kg.

These visualizations help examine how landing success varies by launch site, payload mass, and booster version category.

Predictive analysis (Classification)



Result

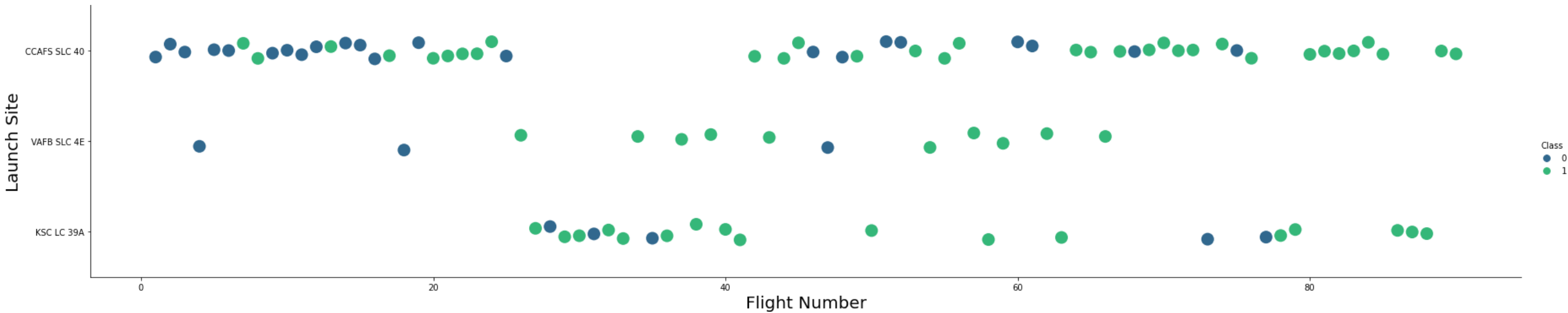


This slide presents a preview of the Plotly interactive dashboard. The following slides will showcase the results of exploratory data analysis through visualizations, SQL-based analysis, geospatial insights using Folium maps, and the final machine learning model results, which achieved approximately 83% accuracy.

EDA with Visualization

EXPLORATORY DATA ANALYSIS WITH SEABORN PLOTS

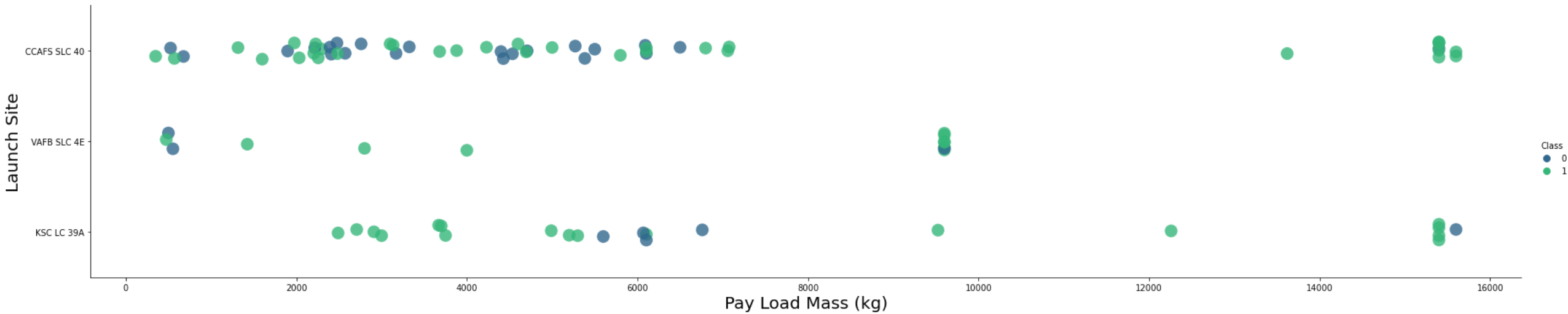
Flight Number vs. Launch Site



Green indicates successful launch; Purple indicates unsuccessful launch.

The visualization indicates that the landing success rate improved over time, as reflected by increasing flight numbers, suggesting operational learning and performance improvements. A notable improvement appears around Flight 20, where the success rate rises significantly. CCAFS also stands out as the primary launch site, handling the highest number of launches.

Payload vs. Launch Site

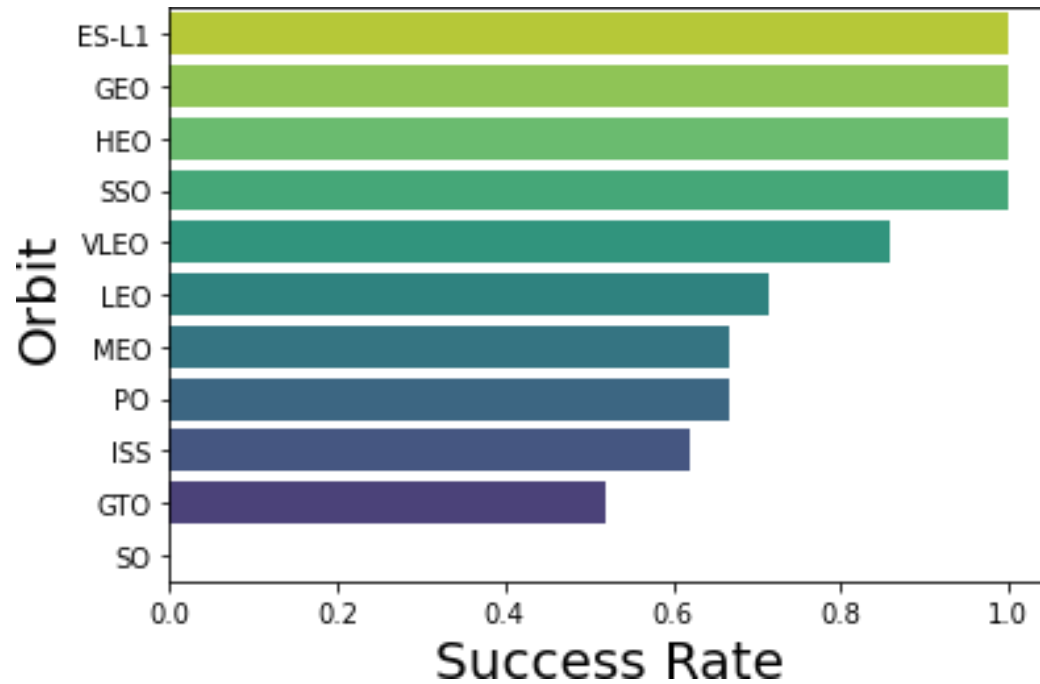


Green indicates successful launch; Purple indicates unsuccessful launch.

Payload mass appears to fall mostly between 0-6000 kg.

Different launch sites also seem to use different payload mass.

Success rate vs. Orbit type



Success Rate Scale with
0 as 0%
0.6 as 60%
1 as 100%

ES-L1 (1), GEO (1), HEO (1) have 100% success rate (sample sizes in parenthesis)

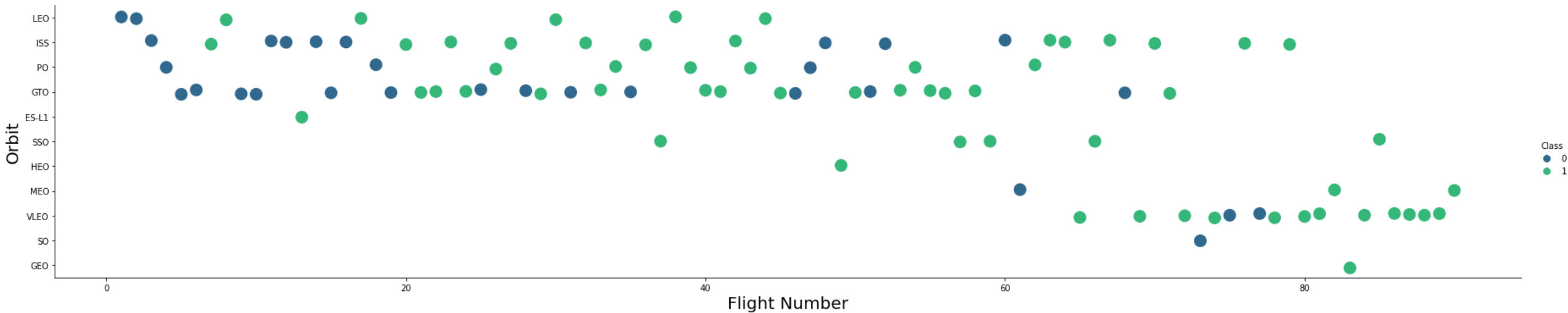
SSO (5) has 100% success rate

VLEO (14) has decent success rate and attempts

SO (1) has 0% success rate

GTO (27) has the around 50% success rate but largest sample

Flight Number vs. Orbit type



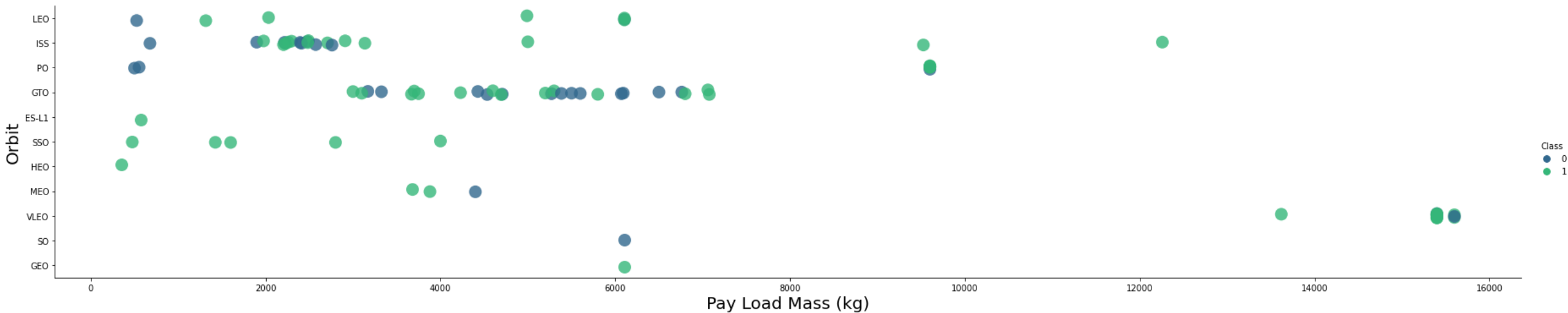
Green indicates successful launch; Purple indicates unsuccessful launch.

Launch Orbit preferences changed over Flight Number. Launch Outcome seems to correlate with this preference.

SpaceX started with LEO orbits which saw moderate success LEO and returned to VLEO in recent launches

SpaceX appears to perform better in lower orbits or Sun-synchronous orbits

Payload vs. Orbit type



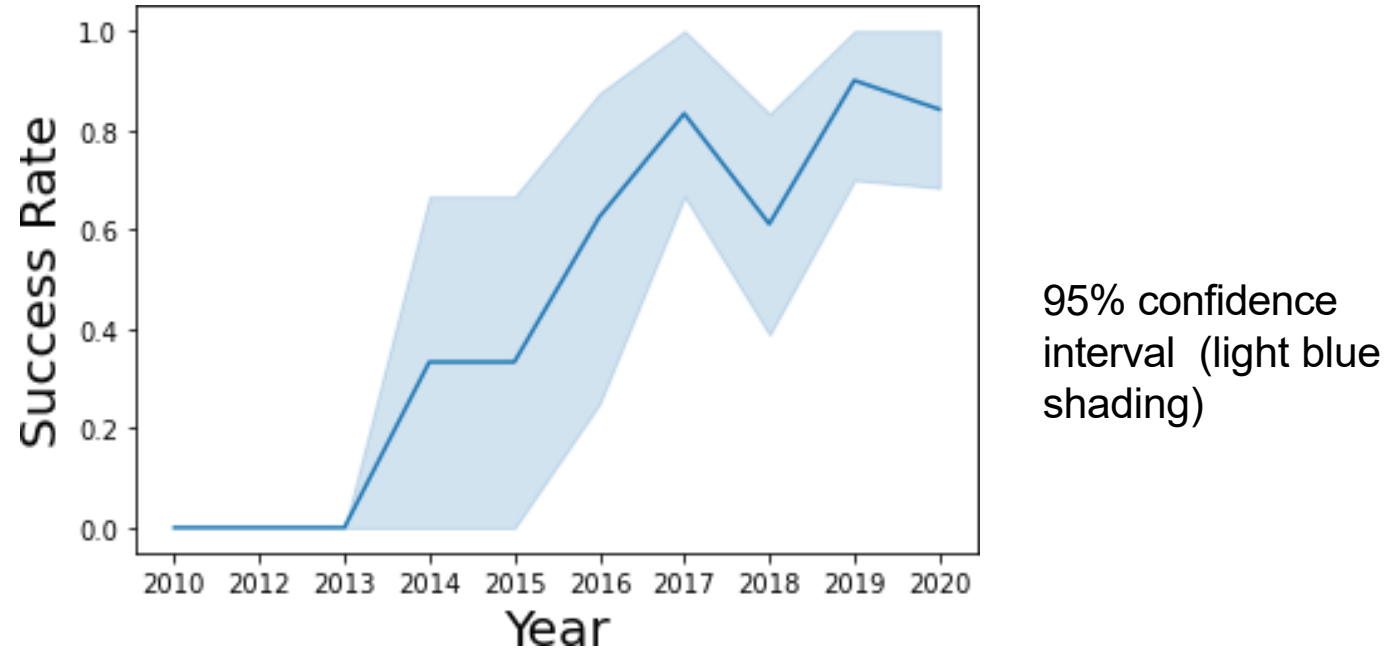
Green indicates successful launch; Purple indicates unsuccessful launch.

Payload mass seems to correlate with orbit

LEO and SSO seem to have relatively low payload mass

The other most successful orbit VLEO only has payload mass values in the higher end of the range

Launch Success Yearly Trend



Success generally increases over time since 2013 with a slight dip in 2018
Success in recent years at around 80%

EDA with SQL

EXPLORATORY DATA ANALYSIS WITH SQL DB2
INTEGRATED IN PYTHON WITH SQLALCHEMY

All Launch Site Names

In [4]:

```
%%sql
SELECT UNIQUE LAUNCH_SITE
FROM SPACEXDATASET;

* ibm_db_sa://ftb12020:***@0c77d6f:
Done.
```

Out[4]:

launch_site
CCAFS LC-40
CCAFS SLC-40
CCAFSSLC-40
KSC LC-39A
VAFB SLC-4E

Query unique launch site names from database.

CCAFS SLC-40 and CCAFSSLC-40 likely all represent the same launch site with data entry errors.

CCAFS LC-40 was the previous name.

Likely only 3 unique launch_site values:

CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

Launch Site Names Beginning with `CCA`

```
In [5]: %%sql
SELECT *
FROM SPACEXDATASET
WHERE LAUNCH_SITE LIKE 'CCA%'
LIMIT 5;
```

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg.databases.appdomain.cloud:31198/bludb
Done.

```
Out[5]:
```

DATE	time__utc_	booster_version	launch_site	payload	payload_mass_kg_	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

First five entries in database with Launch Site name beginning with CCA.

Total Payload Mass from NASA

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS SUM_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE CUSTOMER = 'NASA (CRS)';
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86
Done.
```

sum_payload_mass_kg
45596

This query sums the total payload mass in kg where NASA was the customer.

CRS stands for Commercial Resupply Services which indicates that these payloads were sent to the International Space Station (ISS).

Average Payload Mass by F9 v1.1

```
%%sql
SELECT AVG(PAYLOAD_MASS_KG_) AS AVG_PAYLOAD_MASS_KG
FROM SPACEXDATASET
WHERE booster_version = 'F9 v1.1'
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-80
Done.
```

avg_payload_mass_kg

2928

This query calculates the average payload mass of launches which used booster version F9 v1.1

Average payload mass of F9 1.1 is on the low end of our payload mass range

First Successful Ground Pad Landing Date

```
%%sql
SELECT MIN(DATE) AS FIRST_SUCCESS
FROM SPACEXDATASET
WHERE landing__outcome = 'Success (ground pad)';
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81
Done.
```

first_success

2015-12-22

This query returns the first successful ground pad landing date.

First ground pad landing wasn't until the end of 2015.

Successful landings in general appear starting 2014.

Successful Drone Ship Landing with Payload Between 4000 and 6000

```
%%sql
SELECT booster_version
FROM SPACEXDATASET
WHERE landing__outcome = 'Success (drone ship)' AND payload_mass__kg_ BETWEEN 4001 AND 5999;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.database
Done.
```

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

This query returns the four booster versions that had successful drone ship landings and a payload mass between 4000 and 6000 noninclusively.

Total Number of Each Mission Outcome

```
%%sql
SELECT mission_outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
GROUP BY mission_outcome;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-1
Done.
```

mission_outcome	no_outcome
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

This query returns the count of each mission outcome.

The results indicate that SpaceX successfully achieves its mission objectives in nearly 99% of launches, suggesting that most landing failures are planned rather than mission failures.

The data also shows one launch with an unclear payload status and one mission that failed during flight.

Boosters that Carried Maximum Payload

```
%%sql
SELECT booster_version, PAYLOAD_MASS__KG_
FROM SPACEXDATASET
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXDATASET);

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1
Done.
```

booster_version	payload_mass__kg_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

This query returns the booster versions that carried the highest payload mass of 15600 kg.

These booster versions are very similar and all are of the F9 B5 B10xx.x variety.

This likely indicates payload mass correlates with the booster version that is used.

2015 Failed Drone Ship Landing Records

```
%%sql
SELECT MONTHNAME(DATE) AS MONTH, landing__outcome, booster_version, PAYLOAD_MASS__KG_, launch_site
FROM SPACEXDATASET
WHERE landing__outcome = 'Failure (drone ship)' AND YEAR(DATE) = 2015;

* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.app
Done.
```

MONTH	landing__outcome	booster_version	payload_mass__kg_	launch_site
January	Failure (drone ship)	F9 v1.1 B1012	2395	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	1898	CCAFS LC-40

This query returns the Month, Landing Outcome, Booster Version, Payload Mass (kg), and Launch site of 2015 launches where stage 1 failed to land on a drone ship.

There were two such occurrences.

Ranking Counts of Successful Landings Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT landing__outcome, COUNT(*) AS no_outcome
FROM SPACEXDATASET
WHERE landing__outcome LIKE 'Succes%' AND DATE BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY landing__outcome
ORDER BY no_outcome DESC;
```

```
* ibm_db_sa://ftb12020:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg
Done.
```

landing__outcome	no_outcome
Success (drone ship)	5
Success (ground pad)	3

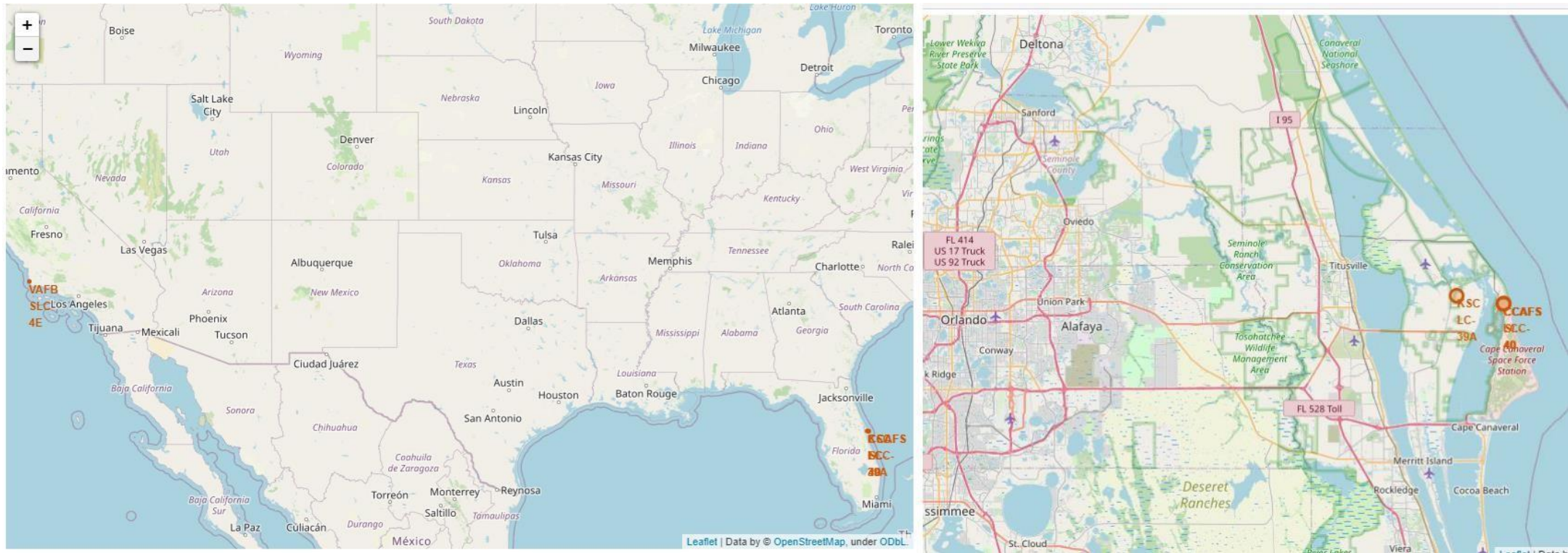
This query returns a list of successful landings and between 2010-06-04 and 2017-03-20 inclusively.

There are two types of successful landing outcomes: drone ship and ground pad landings.

There were 8 successful landings in total during this time period

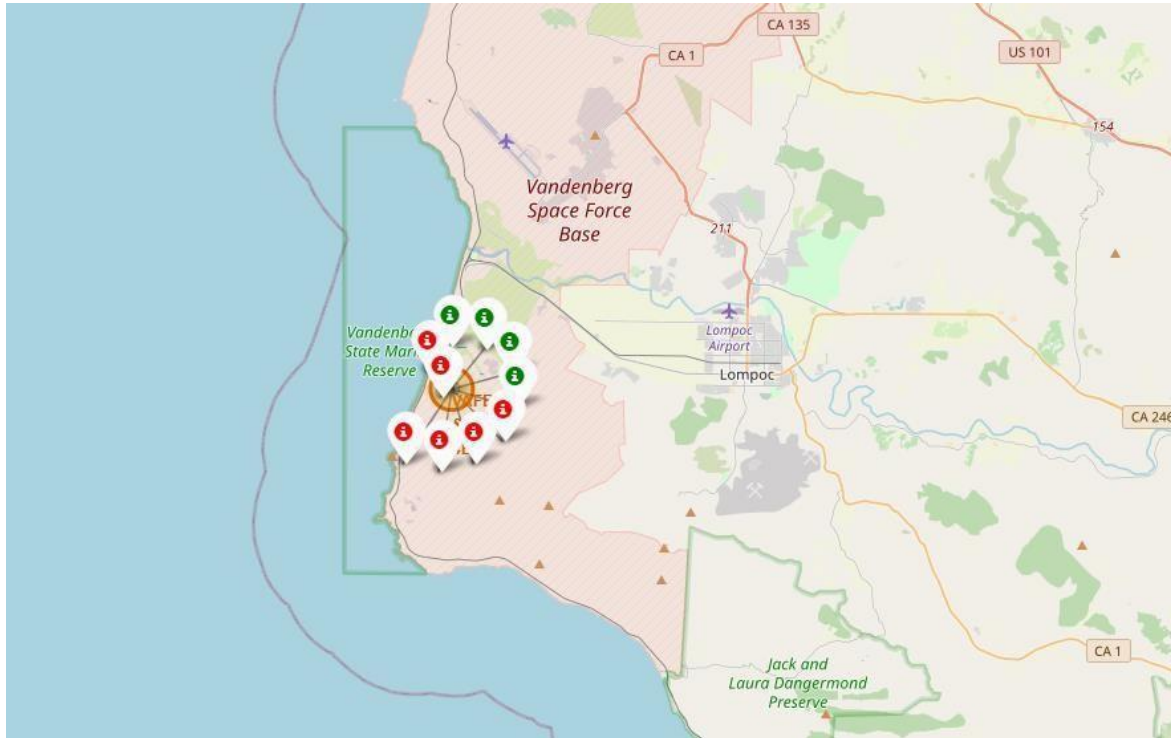
Interactive Map with Folium

Launch Site Locations



The left map displays all launch sites across the United States, while the right map provides a closer view of the two Florida launch sites due to their close proximity. Notably, all launch sites are strategically located near the coastline.

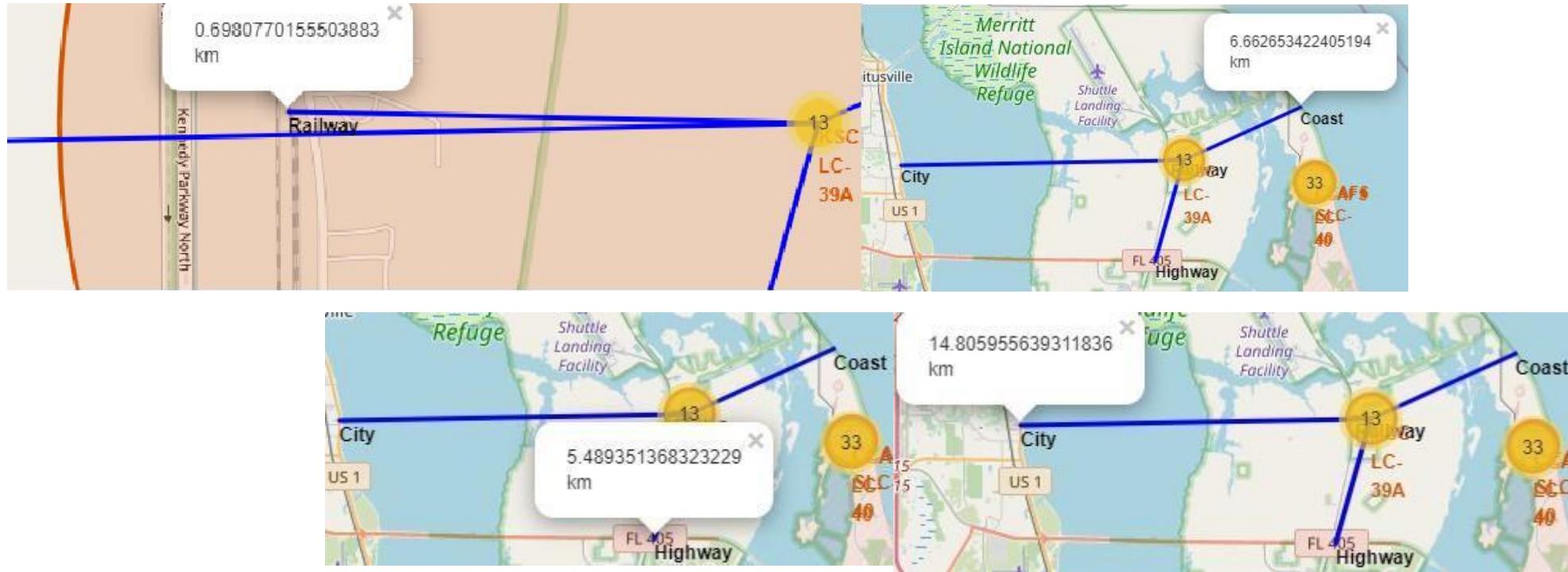
Color-Coded Launch Markers



Clusters on Folium map can be clicked on to display each successful landing (green icon) and failed landing (red icon).

In this example VAFB SLC-4E shows 4 successful landings and 6 failed landings.

Key Location Proximities



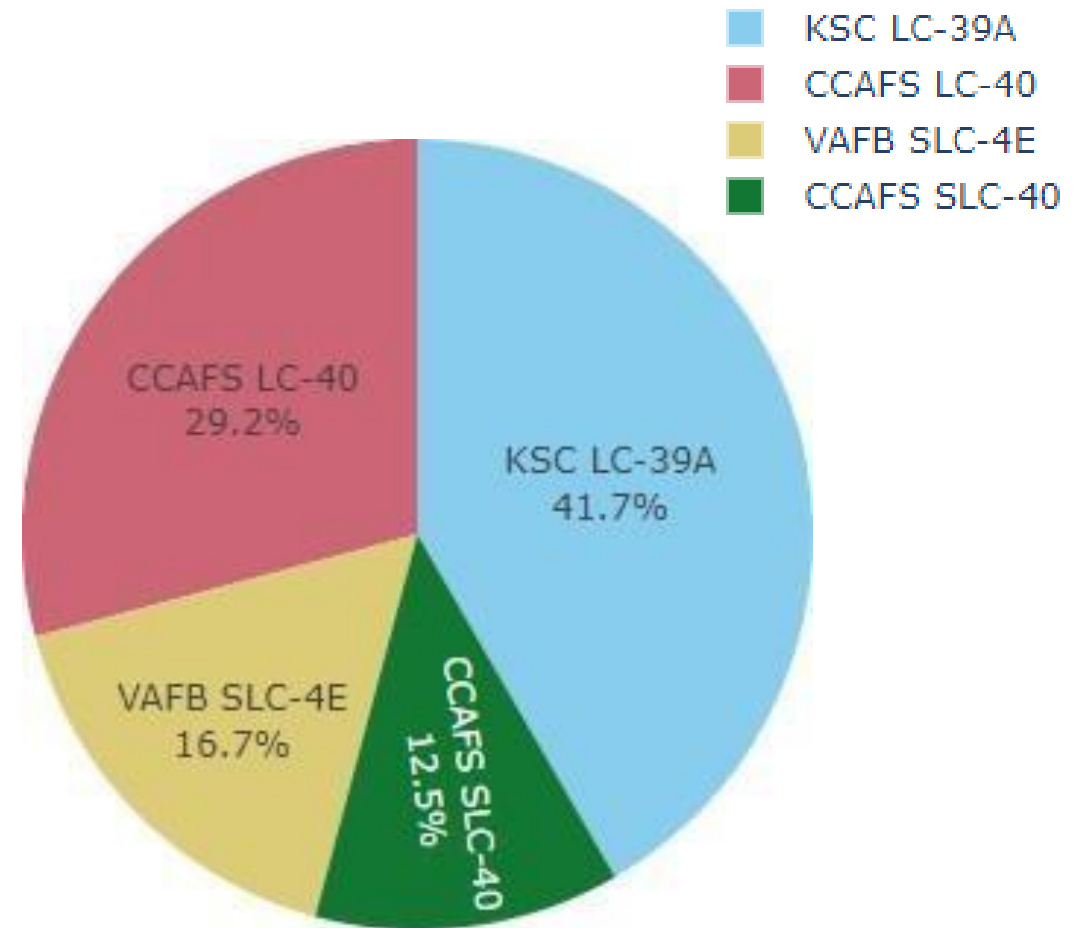
Using KSC LC-39A as an example, launch sites are strategically positioned near railways to support the transportation of large rocket components and supplies, and close to highways for personnel and logistical movement. They are also located near the coastline and at a safe distance from major cities, allowing failed launches to fall into the ocean rather than posing risks to densely populated areas.

Build a Dashboard with Plotly Dash

Successful Launches Across Launch Sites

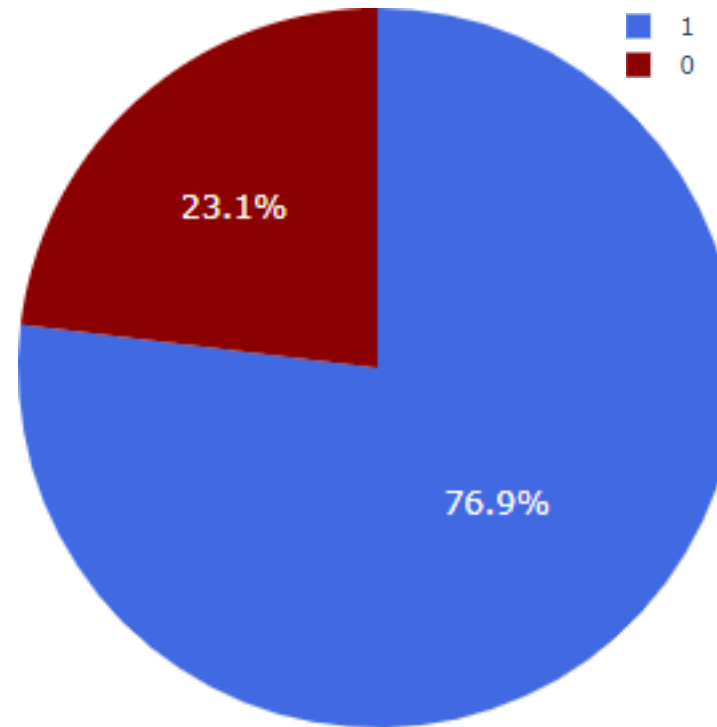
This chart shows the distribution of successful landings across all launch sites. CCAFS LC-40, the former name of CCAFS SLC-40, and KSC show similar numbers of successful landings, although most of the successes occurred before the site name change.

VAFB has the smallest share of successful landings, which may be due to a smaller number of launches and the added operational challenges associated with west coast missions.



Highest Success Rate Launch Site

KSC LC-39A has the highest success rate with 10 successful landings and 3 failed landings.



KSC LC-39A Success Rate (blue=success)

Payload Mass vs. Success vs. Booster Version Category



The Plotly dashboard includes a payload range selector, configured from 0 to 10,000 kg, although the maximum payload in the dataset is approximately 15,600 kg.

The target variable class represents landing outcomes, where 1 indicates a successful landing and 0 indicates a failure.

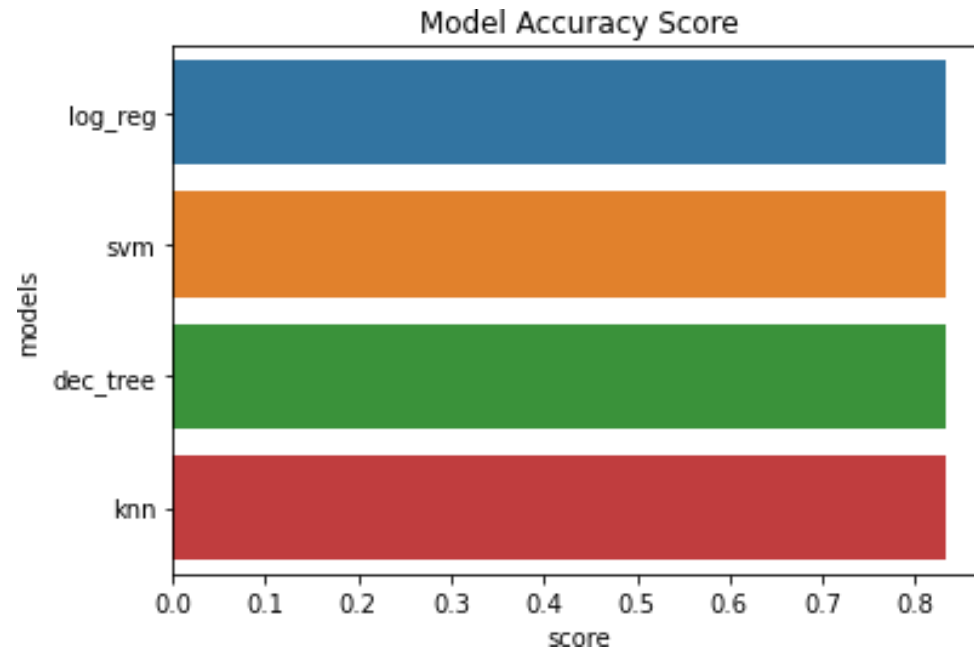
The scatter plot also incorporates booster version categories through color coding and launch frequency through marker size.

Within the selected 0–6,000 kg range, an interesting observation is the presence of two failed landings with payloads of 0 kg.

Predictive Analysis (Classification)

GRIDSEARCHCV (CV=10) ON LOGISTIC REGRESSION, SVM, DECISION TREE, AND KNN

Classification Accuracy



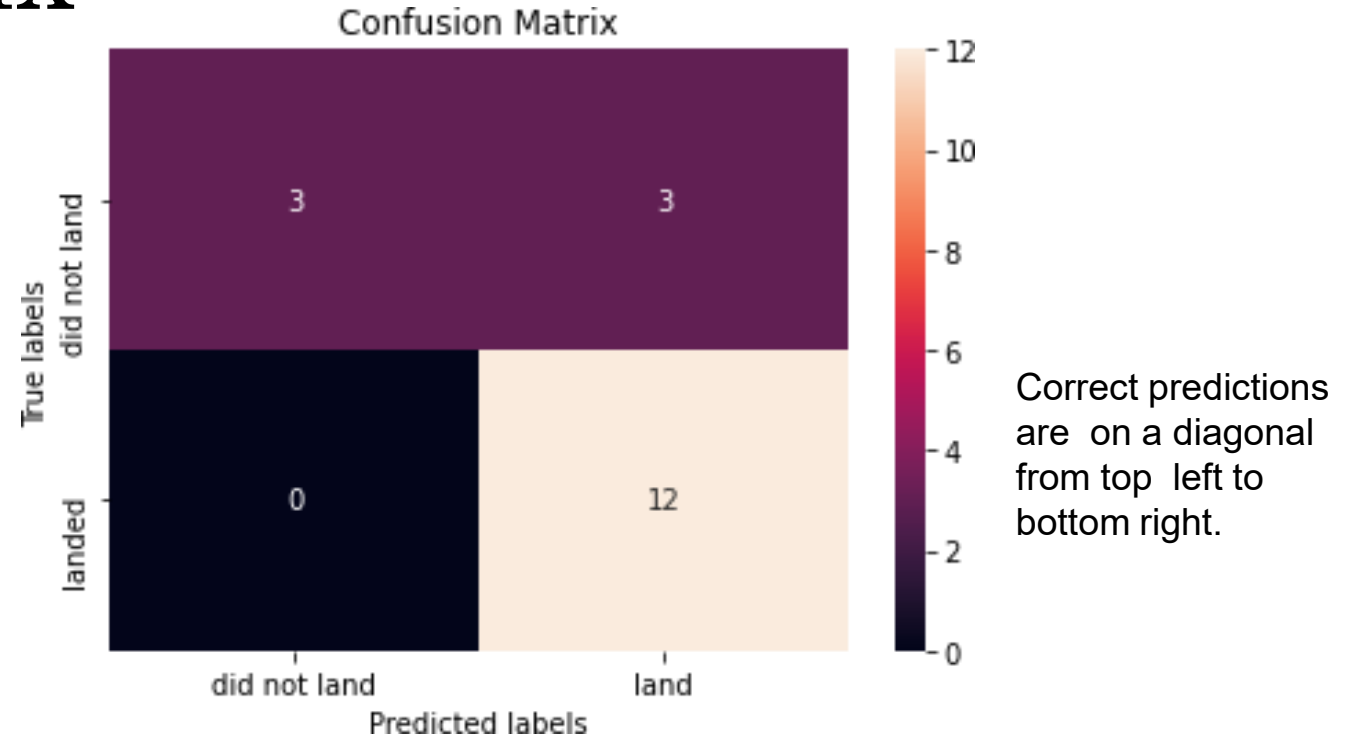
All models had virtually the same accuracy on the test set at 83.33% accuracy.

It should be noted that test size is small at only sample size of 18.

This can cause large variance in accuracy results, such as those in Decision Tree Classifier model in repeated runs.

We likely need more data to determine the best model.

Confusion Matrix



Since all models performed the same for the test set, the confusion matrix is the same across all models. The models predicted 12 successful landings when the true label was successful landing.

The models predicted 3 unsuccessful landings when the true label was unsuccessful landing. The models predicted 3 successful landings when the true label was unsuccessful landings (false positives). Our models over predict successful landings.

CONCLUSION

Our objective was to develop a machine learning model for Space Y, a competitor seeking to bid against SpaceX in the commercial launch market. The goal was to predict the likelihood of a successful first-stage booster landing, which could help reduce launch costs by approximately \$100 million through booster recovery and reuse.

To achieve this, launch data was collected from the public SpaceX API and web-scraped from the SpaceX Wikipedia page. The data was cleaned, labeled, and stored in an IBM DB2 SQL database, followed by analysis through visualizations, dashboards, and predictive modeling.

The final machine learning models achieved an accuracy of approximately 83%, providing a reasonably strong basis for predicting landing success before launch. This model could support strategic launch decision-making for Space Y. However, incorporating additional data would likely improve model reliability and help identify the most effective predictive approach.